

IoT-Based Heart Attack Pattern Detection Using Deep Learning

Research Objectives, Novelty, and Outcomes

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Purpose of this document

This document revises three parts of the thesis following the supervisor's comments. Each research objective now states **exactly one task**. The **novelty** and the **outcomes** of the study are stated explicitly as separate sections, instead of being implied inside other text. For comparison, the text as originally written in the thesis is quoted first in each section.

Research aim (unchanged, thesis Section 1.3)

The aim of this research is to design, build, and test a cheap IoT system that uses deep learning to find heart attack patterns from a single-lead ECG signal, and to show the heart risk to the user in real time on a web page.

1. Research Objectives

1.1 As originally written (thesis Section 1.4)

- 1. To read and study past research on ECG deep learning, single-lead heart attack detection, and IoT health systems, and to find the research gap.*
- 2. To build a low-cost ECG device using an ESP32 chip and an AD8232 sensor that sends data over WiFi.*
- 3. To build and train a 1D convolutional neural network (1D CNN) using the public PTB-XL ECG dataset.*
- 4. To build a Flask web app that takes the ECG data, runs the model, and shows the heart risk in real time.*
- 5. To test the system using accuracy, precision, recall, and F1-score, and to measure how fast it works.*

1.2 Revised objectives (one task each)

To achieve the aim, the study pursues the following objectives. **Each objective states one task only.**

No.	Objective (one task each)	Phase
O1	To review existing research on deep-learning ECG analysis, single-lead myocardial infarction detection, and IoT-based ECG monitoring.	Literature
O2	To identify the research gap addressed by this study.	Literature
O3	To prepare a Lead-I, Normal-versus-MI dataset from the PTB-XL database using patient-disjoint split.	Data and model
O4	To design a 1D convolutional neural network (1D CNN) architecture for single-lead ECG classification.	Data and model
O5	To train the 1D CNN on the prepared training set.	Data and model
O6	To determine the classification threshold on the validation set according to clinical priority (maximising recall).	Data and model
O7	To evaluate the trained model on the unseen test set.	Data and model
O8	To construct a low-cost single-lead ECG acquisition device using an ESP32 microcontroller and an AD8232 sensor.	Hardware
O9	To develop a Flask web application that performs model inference and displays the heart-risk result.	System
O10	To measure the end-to-end latency of the complete system.	System

1.3 How the original objectives map to the revised ones

Original	Contained	Becomes
Original 1	Two tasks: reviewing past work, and finding the gap	O1, O2

Original	Contained	Becomes
Original 2	One task	O8
Original 3	Three tasks: preparing data, designing the network, training it	O3, O4, O5
Original 4	One task	O9
Original 5	Two tasks: measuring detection quality, and measuring speed	O7, O10
(not listed)	Selecting the decision threshold - work carried out but not stated as an objective	O6

Why the change was needed. Several original objectives combined two or three tasks in one sentence, so a single objective produced more than one result and could not be marked as met or not met. Original objective 5, for example, asked both to *test the system using accuracy, precision, recall and F1-score* and to *measure how fast it works* — two different activities. Preparing the dataset and selecting the decision threshold were substantial pieces of work that were carried out but were not listed as objectives at all.

2. Novelty of the Study

2.1 As originally written (thesis Section 2.5. Research Gap)

Putting all of this together, the research gap that this project fills is the lack of one full, cheap, single-lead system that uses deep learning to find heart attack patterns in real time. Past work has shown, on its own, that deep learning can read ECGs very well, that single-lead heart attack detection is possible, and that IoT hardware can record and send ECG signals cheaply. But no well-known study has joined these three things into one full system that is (i) made just for heart attacks, (ii) built on cheap single-lead hardware, and (iii) tested for speed as well as accuracy. This project fills that gap by building and testing exactly this kind of system.

2.2 Novelty stated explicitly

The novelty of this research does not lie in inventing a new neural network. It lies in **combining three elements that previous work has kept separate**, and in evaluating the result under the constraints of low-cost hardware. The novelty is stated below as four specific claims.

No.	Novelty claim	Why it is new
N1	A complete, working pipeline from a low-cost single-lead sensor, through ECG data acquisition, to deep learning and displaying and signal, leaving	But, the ECG data is usually noisy, and deep learning and displaying and signal, leaving
N2	The system targets myocardial infarction specifically, rather than simply trying to detect any abnormality.	Existing systems that do add intelligence detect arrhythmia or atrial fibrillation
N3	The model is trained on Lead I only, because Lead I is the lead most widely recorded by a single-lead device.	Most single-lead ECG devices record only a single lead, and their results are as
N4	The decision threshold is selected on the validation set to prioritize speed over accuracy.	Most single-lead systems are defined at a 0.5 threshold for operating point, mis

Stating the limits of the novelty honestly. The individual components — the 1D CNN, the PTB-XL dataset, the AD8232 sensor — are all established and are not claimed as new. The contribution is the *integration*, the *MI-specific single-lead focus*, and the *reporting of speed alongside accuracy*. Claiming more than this would not survive examination.

3. Outcomes of the Study

The outcomes are of two kinds: the **artefacts** the project delivers, and the **measured results** that answer the research questions.

3.1 Artefacts delivered

No.	Artefact
A1	A prepared single-lead dataset of 14,538 Lead-I recordings (9,069 Normal and 5,469 MI), split into patient-disjoint training, validation and
A2	A trained 1D CNN model of 229,473 parameters, released as an ensemble of five models.
A3	A Flask web application that performs model inference and displays the ECG trace together with the predicted heart-risk result.
A4	A low-cost single-lead ECG acquisition device built from an ESP32 microcontroller and an AD8232 sensor.
A5	An integrated end-to-end system joining the acquisition device to the web application over Wi-Fi.

3.2 Measured results

The following results were obtained on the **test set (fold 10, 1,462 recordings)**, which was not used at any point during training or tuning.

Measure	Result	Meaning
ROC-AUC	0.9076	How well the two classes separate overall
PR-AUC	0.8942	The same, but fairer when classes are uneven

Measure	Result	Meaning
Recall (sensitivity)	0.9145	503 of 550 real heart attacks were detected
Specificity	0.6678	609 of 912 healthy records were left alone
Precision	0.6241	Of all MI alarms raised, 62.4% were correct
F1-score	0.7419	Balance of precision and recall
Accuracy	0.7606	Proportion of all predictions that were correct
Missed heart attacks	47	False negatives - the most costly error
Server inference time	about 0.35 s	Model prediction time on the server

Measured at the recall-first operating point (threshold 0.2974, selected on the validation set).

An ensemble of five models improved ROC-AUC from 0.9017 to 0.9076. The difference is +0.0059 with a 95% confidence interval of [+0.0009, +0.0110]. Because this interval does not include zero, the improvement is statistically significant.

3.3 Outcomes answering the research questions

RQ	Outcome
RQ1	The signal quality obtained from the ESP32 and AD8232 acquisition device, assessed against the dataset recordings.
RQ2	A single-lead deep-learning model that detects MI patterns with ROC-AUC 0.9076 and recall 0.9145 on unseen data.
RQ3	The end-to-end latency of the system, from device transmission to the result being displayed on the web dashboard.

4. Mapping: Objective → Research Question → Outcome

This table shows that every objective produces a stated outcome, and that every research question is answered by at least one objective.

Objective	RQ	Outcome	Where shown
O1 Review literature	—	Research context established	Chapter 2
O2 Identify gap	—	Research gap stated	Section 2.5
O3 Prepare dataset	RQ2	A1 - 14,538 Lead-I records	Chapter 4
O4 Design 1D CNN	RQ2	A2 - model architecture	Chapter 4
O5 Train the model	RQ2	A2 - trained model	Chapter 4
O6 Select threshold	RQ2	Threshold 0.2974, recall-first	Chapter 5
O7 Evaluate the model	RQ2	ROC-AUC 0.9076, recall 0.9145	Chapter 5
O8 Build the device	RQ1	A4 - ESP32 + AD8232 prototype	Chapter 4
O9 Build the web app	RQ3	A3 - Flask application	Chapter 4
O10 Measure latency	RQ3	A5 - end-to-end latency	Chapter 5

5. Related corrections needed elsewhere in the thesis

The revised objectives describe what was actually built. Three statements currently in Chapter 3 do not match the implementation and should be corrected at the same time, so the document stays consistent.

Section	Currently says	Should say
3.5	The data is split into about 70 percent training, 15 percent validation, and 15 percent test.	The official PTB-XL stratified folds are used: folds 1-8 for training (11,622 records), fold 9 for validation (1,162 records), and fold 10 for testing (1,162 records).
3.5	A band-pass filter removes slow drift and high-frequency noise.	Either implement the filter, or state that per-recording z-score normalisation is the only processing step.
3.4 / 2.3	It has 21,837 ten-second records from 18,885 patients.	Version 1.0.3 of PTB-XL, used in this study, contains 21,799 records from 18,869 patients.

One further point to consider stating. Of the 5,469 MI records used, only about 6% carry a code indicating an *acute* infarction; the remainder are prior or healed infarctions. The system therefore detects MI *patterns*, and is best described as a screening and triage tool rather than an acute emergency alarm. Stating this protects the claims made in Sections 1.1 and 1.3.